Hannah Arnold
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I'm a First Class AI and Robotics graduate with a strong interest in embedded software engineering and low-level programming. Through university projects I have developed skills in embedded programming, integrating sensors, motors and peripherals while debugging hardware and software issues. I enjoy solving technical problems and am looking to begin my career in embedded programming or software development.

Key Skills

Programming & Software Development

- C and C++ programming
- Python programming
- Structured programming and algorithm design
- Low-level programming and hardware abstraction
- Git and GitHub
- Technical problem-solving

Embedded Systems & Microcontrollers

- STM32 microcontrollers
- GPIO, timers and basic peripherals
- Sensor and actuator integration
- Hardware and software interfacing
- Breadboard prototyping and external power supplies
- Hardware debugging and fault finding
- Firmware debugging

Additional Professional Skills

- Teamwork and collaboration
- Leadership and organisation
- Communication
- Decision-making
- Adaptability
- Working effectively under pressure

Education

Staffordshire University | September 2023 - May 2026

BSc (Hons) AI and Robotics - First Class Honours

Relevant Modules

<u>Robotic Programming</u>

- Implemented hardware-level programming using GPIO, timers, and basic peripherals
- Developed embedded applications in C for STM32 microcontrollers

<u>Final Year Project</u>

- Designed and developed an embedded system using STM32 Microcontrollers
- Integrated hardware and software components including sensors
- Built and tested hardware interfacing with external power supply, external buttons on a bread board and a stepper motor

AI and Chatbots

- Developed software solutions using Python and C++
- Gained experience in algorithm design and structured programming

City of Stoke-On-Trent Sixth Form College | September 2021 – July 2023
Grade range: B - D
- Mathematics
- Physics
- Further Mathematics
- Extended Project Qualification (EPQ)

Projects
Final Year Project - Implementing microcontrollers and sensors to determine ride path on roller coasters October 2025 - May 2026
Developed an embedded system exploring how roller-coaster user experience could be gamified without virtual reality. The system used multiple sensors and a camera to collect rider data, which was used to determine emotion and select a suitable ride path.

- Developed embedded systems in C using STM32 Microcontrollers
- Interfaced multiple sensors
- Debugged hardware and firmware issues
- Designed and tested electronic circuits

Study Progression Website May 2026 - Present

Portfolio: xxhansterxx.github.io

Developing a personal portfolio to document ongoing technical development and showcase projects across web development, embedded programming and software engineering.

- Built and maintained a personal portfolio website.
- Published projects and experiments involving semantic HTML, embedded programming and more
- Continuously developing technical skills through independent study and practical projects.

Employment
Retail Assistant - Primark | April 2022 - Present
- Deliver excellent customer service in a fast-paced retail environment.
- Demonstrate strong communication, teamwork and adaptability.
- Trusted to work across multiple departments and tills while maintaining accuracy.
- Manage changing priorities effectively in a busy working environment.

Hobbies
Macclesfield Warriors Hockey Team - Committee Member & Captain | December 2024 – Present
- Lead a competitive team during training and matches.
- Organise team communication and fixtures as part of the committee.
- Develop leadership, organisation and decision-making skills.
- Handle conflicts calmly and effectively in high-pressure situations.

University Engineering Society | September 2023 – May 2024
- Participated in practical electronics and engineering workshops.
- Gained hands-on experience soldering LEDs.
- Used CAD software for 3D printing and laser-cutting projects.